MODE OF TRANSACTION

- Lecture, Seminar, Discussion

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses with 2 Credit Theory + 2 Credit Practical.

Sample Questions to test Outcomes

1. Explain the difference between JDK, JRE, and JVM. How do they contribute to the execution of Java programs?
2. Discuss the role of try-catch blocks in Java exception handling. Provide an example scenario where you would use each of the following: throw, throws, and finally.
3. What is synchronization in Java multithreading? Why is it necessary? Explain the life cycle of a thread in Java.
4. Describe the steps involved in establishing database connectivity using JDBC with MySQL

KU5DSCCSE303 COMPUTER NETWORK

Semester	Course Type	Course Level	Course Code	Credits	Total Hours
5	DSC	300	KU5DSCCSE303	4	90

Learning Approach (Hours/ Week)			Marks Distribution			Duration of ESE (Hours)
Lecture	Practical/ Internship	Tutorial	CE	ESE	Total	
2	4	1	50	50	100	2(T)+3(P)*

*** ESE Duration : 2 hours for theory and 3 hours for Lab**

Course Description:

This course exposes the fundamental concepts of main concepts of networking to the learner. TCP/IP network model is taken as the basis for the discussion. The course provides the learner a brisk walk through the fundamental concepts of application layer, transport layer, data link layer, and transport layer.

Course Objectives:

- Provide basic knowledge of the concepts of computer networks
- Give a basic idea about the OSI Model and TCP/IP Network models
- Familiarize with the basic concepts of various layers in TCP/IP network model

Course Outcomes:

At the end of the Course, the Student will be able to:

SL #	Course Outcomes
CO1	Explain the features of computer networks, protocols, and network design models
CO2	Illustrate the functions and various algorithms / protocols used of application layer and transport layer of TCP/IP model
CO3	Explain the functions and various algorithms / protocols used of network layer of TCP/IP model
CO4	Describe the functions and various algorithms / protocols used of data link layer of TCP/IP model

Mapping of COs to PSOs

	PSO1	PSO2	PSO3	PSO4	PSO5
CO1	✓	✓	✓	✓	✓
CO2	✓	✓	✓	✓	✓
CO3	✓	✓	✓	✓	✓
CO4	✓	✓	✓	✓	✓

COURSE CONTENTS

Module 1: Internet: Description - Protocol - Network Edge - Network Core. Performance Metrics: Delay - Loss - Throughput. Layered Architecture - The Internet protocol stack - OSI reference model. Application Layer: Network

Application Architectures - Communication of Processes - Services for Transport - Application Layer Protocols. Web and HTTP: HTTP - Persistent and Non Persistent - HTTP Message Format. Socket Programming with UDP and TCP. (20 hours)

Module 2: Transport Layer: Services - Relationship between Transport and Data Link Layer - Multiplexing and Demultiplexing. UDP - Segment Structure - Checksum - Principles of Reliable Data Transfer - GBN - SR. TCP: Connection - Segment Structure - Segment and Acknowledgement Numbers - Reliable Data Transfer Flow Control - Congestion Control. (20 hours).

Module 3: Network Layer: Forwarded Routing - Service Models - VC and Datagram Networks - Functioning of a Router. IP - IPV4 Addressing - Interface Address - Subnet - Netmask - CIDR - ICMP - IPV6 Addressing - Datagram Format - IP V4 to V6 Transition. Routing Algorithms: LS - DV. Intra AS Routing in Internet: RIP - OSPF. (20 hours)

Module 4: Data Link Layer: Services - NIC. Error detection and Correction: CheckSum - Parity - CRC. Multiple Access Links and Protocols: Channel Partitioning Protocol - Random Access Protocols: ALOHA - CSMA - CSMA/CD. Link Layer Addressing: MAC Address - ARP. Ethernet: Frame Structure - Technologies. Link Layer Switches Vs Routers. (20 hours)

Module X (Teacher Specific): Cookies - SMTP - RTT Estimation - Inter AS Routing on the Internet: BGP - IEEE 802.3 - Case Study: A Day in the Life of a Web Page Request - Data Centre Networking. (10 hours)

Core Compulsory Readings

1. James F. Kurose, Keith W. Ross, Computer Networking: A Top-Down Approach, 6th Edition.

Core Suggested Readings

1. Gerry Howser, Computer Networks and the Internet: A Hands-On Approach, Springer, ISBN-13: 9783030344955, 2019.
2. A. Tanenbaum and D. Wetherall, Computer Networks, 5th edition, Pearson, ISBN-13: 9780132126953, 2013.

TEACHING LEARNING STRATEGIES

- Lecturing, Visualization, Team Learning

MODE OF TRANSACTION

- Lecture, Seminar, Discussion, Questioning and Answering

ASSESSMENT RUBRICS

Refer to section 7 of FYIMP- Computational Science - Scheme and Syllabus for the 4 credit courses with 2 Credit Theory + 2 Credit Practical.

Sample Questions to test Outcomes.

1. Define simplex, half-duplex, and full-duplex transmission modes.
2. "Data link protocols almost always put the CRC in a trailer rather than in a header". Justify.
3. Compute the number of octets the smallest possible IPv6 (IP version 6) datagram ?
4. Distinguish the features of UDP and TCP
5. Illustrate the concept of netmask with a suitable example
6. Explain any one random access protocol.